RVMC Clinical Operations

GPC Clinical Corner

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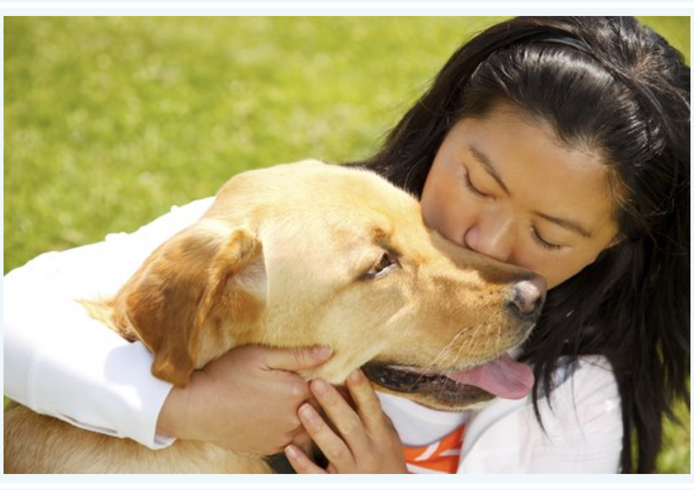
Animal Bites (Dogs, Cats, and Other Mammals)

The Prominence of Animal Bites

Dog bites: These account for approximately 90 percent of animal bites (rate of 103 to 118 per 100,000 population) in the United States. Most dog bite victims are children, with the highest number in boys between 5 and 9 years old. In resource-limited regions, bites by stray dogs are more common causes of rabies transmission than bites from any other animal.

Cat bites: These account for approximately 10 percent of animal bite wounds in the United States.



Evaluating Bites

Obtaining a History

We ascertain the patient's symptoms, including any indicators of infection (e.g., fever); neurovascular compromise (e.g., weakness, numbness, excessive bleeding); alteration in function (e.g., decreased range of motion); and risk for wound infection, rabies, and tetanus.

Note. From *Swollen Hand with Cat Bite* [Photograph], by Australiawide First Aid, n.d., Australiawide First Aid (https://www.australiawidefirstaid.com.au/resources/first-aid-for-a-cat-bite)

Dog bites are associated with a range of injuries, from minor wounds (e.g., scratches, abrasions) to major wounds (e.g., gaping lacerations, deep puncture wounds, tissue avulsions, and crush injuries). The jaws of large dogs, such as pit bull terriers, German shepherds, and Rottweilers, can exert a strong force that may inflict serious injury and damage underlying structures.

Cat bites tend to penetrate deeply with higher risk of deep infection (abscess, septic arthritis, osteomyelitis, tenosynovitis, bacteremia, or necrotizing soft tissue infection) than dog bites.

Delayed presentation for care (≥12 hours after a bite on the extremities and ≥24 hours after a bite on the face) increases risk of infection.

Medical Conditions that Increase Risk of Infection

Information about the patient

- Date of last tetanus vaccine.
- Dates of prior rabies vaccinations or exposures.
- Geographic location at the time of the bite (rabies is rare in the United States but more common in other parts of the world).

Information about the animal

- Whether the bite was provoked or unprovoked.
- Whether the animal is wild, domestic, or stray.

Current location of animal

- Whether the animal is still alive and available for observation.
- Rabies vaccination history of animal.
- Health status of animal: Recent behavioral changes (e.g., confusion, unsteady gate, aggression, or foaming at the mouth).

Physical Exam

Examine the wound for proximity to important structures (e.g., joints, tendon, bone), depth, foreign material, cosmetic implications, neurovascular status distal to the wound, and signs of infection. Wound proximity to a joint may not be apparent without testing full range of motion.

Infections, if present, include cellulitis, abscess, tenosynovitis, or deep space infection.

Assessing for Infection

Cellulitis

This is an infection of the deep dermis and subcutaneous fat. Clinical manifestations include erythema, tenderness, swelling, and warmth; fever, purulent drainage, or lymphangitis may be present.

Flexor Tenosynovitis

This condition requires prompt recognition and surgical debridement because an infection of the finger flexor tendons can spread rapidly along the sheath compartments into the hand and forearm. Bite wounds to a finger are the most common etiology; the flexor tendons are preferentially involved.

The cardinal signs of flexor tenosynovitis are tenderness along the course of the flexor sheath, symmetric or fusiform enlargement of the affected digit, slight flexion of the digit at rest, and pain along the tendon with passive extension.

Note. From *Flexor tenosynovitis* [Photograph], by UpToDate, n.d., UpToDate (https://www.uptodate.com/contents/image?imageKey=SURG%2F51632&topicKey=ID%2F7671&search=animal%20bite%20treatment&rank=1%7E119&source=see_link)

Necrotizing Soft Tissue Infection (e.g., Fasciitis)

Clinical features of necrotizing soft tissue infections include the following:

- Rapidly progressive pain
- Pain out of proportion to skin exam findings
- Crepitus
- Brawny induration
- Systemic signs such as fever >100.5°F/38°C, hypotension, and/or sustained tachycardia
- Persistent or progressive signs of infection despite initial wound care and appropriate antibiotic administration

Septic Arthritis or Osteomyelitis

These infections can occur if the initial bite penetrated bone or a joint space, or if soft tissue infection spreads to bone or joint.

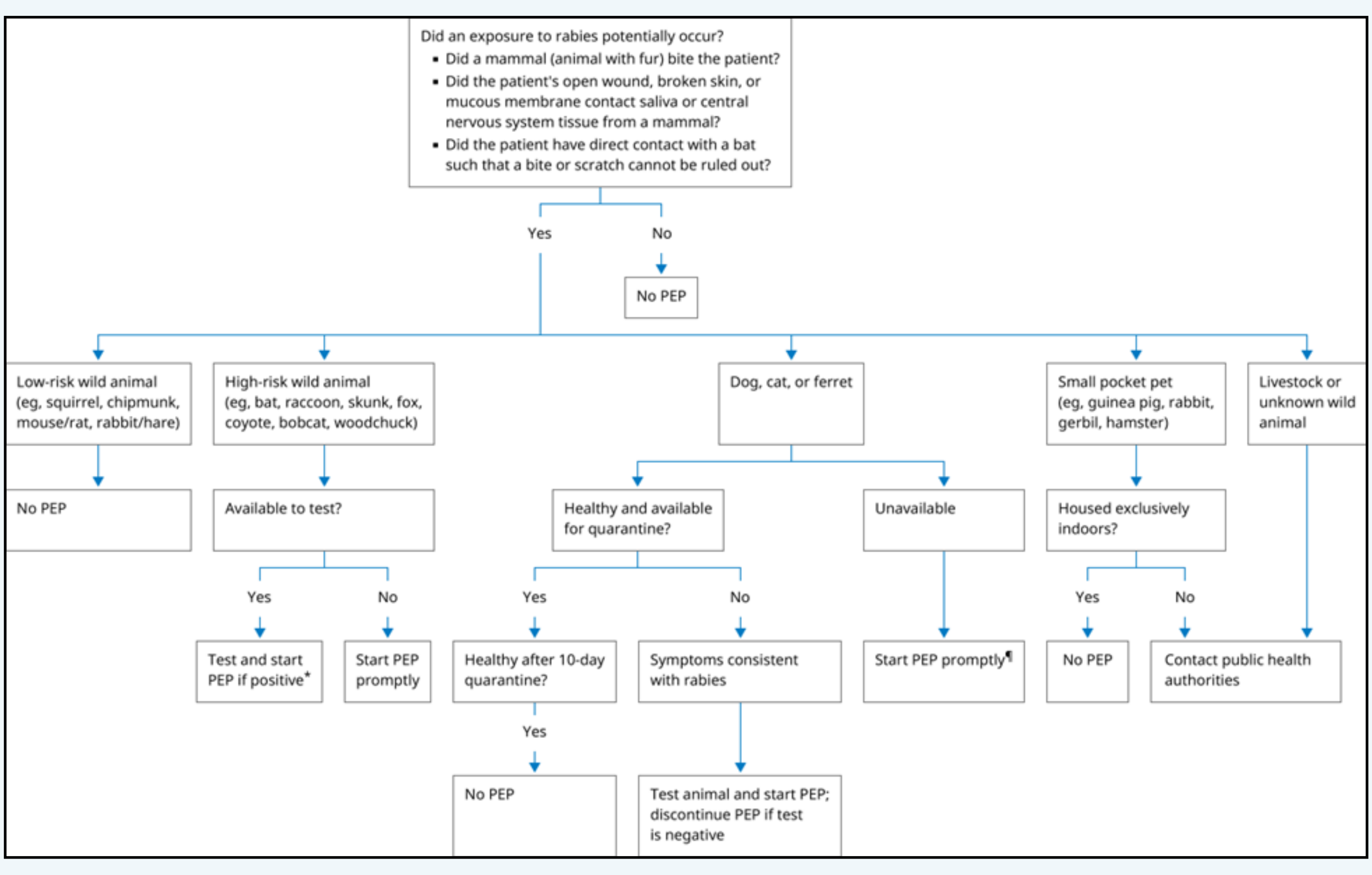
Bite Management

Tetanus

Animal bites are considered a tetanus-prone wound. In a fully vaccinated patient (i.e., has had ≥3 doses of a tetanus toxoid-containing vaccine), administer a tetanus toxoid-containing vaccine only if their last dose was given ≥5 years ago (or if the date of their most recent vaccine is uncertain). In an unvaccinated or partially vaccinated patient, administer a tetanus toxoid-containing vaccine **and** tetanus immune globulin.

Rabies and Tetanus Post-Exposure Prophylaxis

All patients with a mammalian bite require evaluation for rabies and/or tetanus post-exposure prophylaxis. The prevalence of rabies and tetanus varies geographically, but both infections can be fatal if prophylaxis is not provided when indicated.



Infected Dog and Cat Bites

In addition to possible rabies and tetanus prophylaxis, infected bite wounds are managed with antibiotic therapy, obtaining cultures, and wound debridement.

Oral Regimens

For oral treatment of infected dog and cat bites, we suggest one of the following antibiotic regimens:

- **Preferred agent:** Amoxicillin-clavulanate (875/125 mg): Adults BID dosing, Peds (40-45mg/kg/day of amox component divided BID). This regimen covers the most frequently isolated organisms from infected dog and cat bites and is well tolerated in most patients.
- **Alternate regimens:** For a patient unable to take amoxicillin-clavulanate, we suggest combination therapy with one agent that covers aerobic bacteria and another that covers anaerobes.
- For patients intolerant of all beta-lactams, our favored regimen is trimethoprim-sulfamethoxazole (1 double-strength tablet twice daily) plus metronidazole (500 mg orally every 8 hours).

Uninfected Dog and Cat Bites

In a patient with no symptoms or signs of wound infection, management includes wound care and assessment for antibiotic, tetanus, and rabies prophylaxis.

For most patients, we suggest that bite wounds be left open to heal by secondary intention rather than primary closure. Primary closure increases risk of infection and does not significantly improve cosmetic outcomes for most patients.

Prophylactic Antibiotic Regimens

We prefer amoxicillin-clavulanate (875/125 mg every 12 hours in adults) for dog and cat bite prophylaxis. The first dose should be administered as soon as possible after the bite.

Follow-Up Care

All patients with mammalian bites require close follow-up. We examine all wounds two to three days after initial presentation

References

Australiawide First Aid. (n.d.). *Swollen Hand with Cat Bite* [Photograph]. Australiawide First Aid. https://www.australiawidefirstaid.com.au/resources/first-aid-for-a-cat-bite

UpToDate. (n.d.). *Flexor tenosynovitis* [Photograph]. UpToDate. https://www.uptodate.com/contents/image?imageKey=SURG%2F51632&topicKey=ID%2F7671&search=animal%20bite%20treatment&rank=1%7E119&source=see_link